

Activity 2

Chain of Thought – Structured Analysis of a Transaction Failure

Prompt Experiment, Comparison and Key Learning

Scenario: A customer tries to transfer ■50,000, but the transaction fails. The only error shown is “Transaction could not be completed.”

1. Objective

The objective of this activity is to demonstrate how a structured reasoning-oriented prompt can help an AI analyze an ambiguous problem systematically. The activity compares a direct prompt with a prompt that explicitly guides the model to decompose the problem, generate possible causes, check for missed scenarios, and consolidate the findings.

2. Given Scenario

A customer tries to transfer ₹50,000, but the transaction fails. The only error shown to the customer is:

“Transaction could not be completed.”

Expected discoveries from the activity:

- Daily limit exceeded
- Insufficient balance
- Bank server unavailable
- OTP expired
- Network timeout
- Beneficiary inactive
- Fraud rule triggered
- Core banking unavailable
- UPI switch unavailable

3. Experiment 1 – Standard Prompt

Prompt used:

A customer tries to transfer ■50,000, but the transaction fails. The only error shown to the customer is:

"Transaction could not be completed."

What could be the possible reasons for this transaction failure?

Observed result

The model produced a broad list of possible causes covering balance, transfer limits, beneficiary issues, bank/server availability, network failures, authentication, fraud/security, KYC/account restrictions, API/backend and other system-related conditions.

Result: The standard prompt identified all nine expected discoveries.

4. Experiment 2 – Chain-of-Thought-Oriented Prompt

Prompt used:

A customer tries to transfer ■50,000, but the transaction fails. The only error shown to the customer is:

"Transaction could not be completed."

Analyze this problem systematically before providing the final answer.

1. Decompose the transaction into the major stages involved in completing the transfer.
2. For each stage, identify what could potentially cause the transaction to fail.
3. Consider customer-related, business-rule, authentication, system, network, and external-service possibilities.
4. Check whether any important failure scenarios have been missed.
5. Finally, provide a consolidated list of the possible reasons for the transaction failure.

Observed result

The model decomposed the transfer into lifecycle stages: customer initiation, input validation, authentication, account/limit checks, fraud/compliance checks, funds reservation/debit, payment/bank processing, beneficiary-bank processing, and transaction confirmation/final status update.

It then generated failure possibilities at each stage and consolidated them into broader categories. It also checked for overlooked scenarios such as lost confirmation, duplicate retry risk, race conditions and inconsistent transaction status.

5. Decompose → Generate → Debugging Off-Ramp

The second prompt demonstrates the reasoning workflow taught in the session:

Stage	Application in this activity
Decompose	Break the transaction into major lifecycle stages instead of treating the failure as one undifferentiated problem.
Generate	For each stage, generate plausible failure conditions across customer, business-rule, authentication, system, network and external-service areas.
Debugging Off-Ramp	Check whether important scenarios were missed and explore edge cases such as lost confirmation, duplicate retries and inconsistent transaction status.
Consolidate	Bring the findings together into a structured final list of possible reasons.

6. Prompt 1 vs Prompt 2

Aspect	Standard Prompt	CoT-Oriented Prompt
Approach	Direct request for possible reasons.	Explicitly structured analysis.
Decomposition	Not requested.	Transaction broken into stages.
Failure mapping	Broad list of causes.	Causes considered at each stage.
Missed-scenario check	Not explicitly requested.	Explicitly requested.
Edge cases	Limited structure.	Important edge cases surfaced.
Expected 9 discoveries	All identified.	All identified.
Overall result	Useful broad answer.	More systematic and traceable analysis.

7. Important Observation

The generic error message does not establish the actual backend outcome. The scenario only tells us what the customer sees; it does not prove whether the money was actually transferred.

During the structured analysis, the model identified a **possible edge case**: the external bank may process the \$50,000 transfer successfully, but the confirmation response may be lost. The application could then display the generic failure message even though the external transaction succeeded.

This is a possible scenario discovered during analysis, not the confirmed outcome of the given scenario.

The actual outcome would need to be verified using transaction records and relevant service responses before concluding whether the transfer failed, succeeded, or entered an unknown/pending state.

8. Key Learning

- A direct prompt can already produce a useful list of possible causes.
- A reasoning-oriented prompt can provide a more systematic structure for analyzing an ambiguous problem.
- Decomposing a complex flow helps identify where failures can occur.
- Generating possibilities stage-by-stage reduces the chance of overlooking a failure area.
- A debugging/checking step can surface important edge cases.
- The final answer should distinguish confirmed facts from possible scenarios.

9. Conclusion

The exercise demonstrates that Chain-of-Thought-oriented prompting can guide an AI from a broad answer toward a more structured analysis. The standard prompt identified the expected causes, while the structured prompt made the reasoning process more explicit by decomposing the transaction lifecycle, generating failure possibilities for each stage, checking for missed scenarios, and consolidating the results.

The key learning is not simply that one prompt produces more possible answers than another. Rather, structured reasoning guidance can make the analysis more systematic, traceable and useful when the original problem is ambiguous.

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